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GOLF FUNDRAISER PRO

Phase 1

Admin-Run Tournament

Registration · Check-In · Score Entry · Leaderboard · Emails · Landing Page

Golf Scramble Made Easy · Implementation Design Series · v1.0

1. Phase 1 Goal & Scope

Phase 1 delivers a complete tournament management system the organizer can use at a real event. Golfers register online and check in via QR code. The organizer enters scores manually from a tablet. There is no mobile scoring app yet — that is Phase 2. An admin-only leaderboard shows standings. A branded public landing page accepts registrations and donations.

Deliverable: An organizer can run the CLHS 4th Annual Golf & Give end-to-end using Phase 1 alone. Every subsequent phase adds capability without replacing any Phase 1 work.

Timeline	8–10 weeks (2-week infra sprint + 8-week feature sprint)
New Apps	apps/admin (Expo Router web), apps/web (Next.js), apps/api (ASP.NET)
New Packages	packages/ui, packages/shared-types, packages/theme
Database	Full schema created (see Foundation §4). All future-phase tables exist as empty.
Mobile App	NOT in Phase 1. Admin scores from tablet via admin web app.
Public Leaderboard	Admin-only in Phase 1. Public launch in Phase 2.

2. Sprint 1–2: Infrastructure (2 Weeks)

2.1 Monorepo Bootstrap

- npx create-turbo@latest golf-fundraiser-pro — select npm workspaces
- Create apps/api (dotnet new webapi), apps/admin (npx create-expo-app --template tabs), apps/web (npx create-next-app)
- Create packages/ui, packages/shared-types, packages/theme with package.json and tsconfig.json
- Configure turbo.json pipeline: build depends on ^build; lint and type-check run in parallel
- Add .github/workflows/ci.yml — runs turbo lint, type-check, test on every PR

2.2 Database Setup

- Add Docker Compose (see Foundation §8) and confirm postgis/postgis:16-3.4 starts
- Install EF Core: dotnet add package Microsoft.EntityFrameworkCore.Design Npgsql.EntityFrameworkCore.PostgreSQL NetTopologySuite
- Create ApplicationDbContext with all entities (full schema — Foundation §4)
- dotnet ef migrations add Phase1_InitialSchema
- Run: CREATE EXTENSION IF NOT EXISTS postgis; in migration's Up() raw SQL
- Seed: Eco Green default theme, one demo organization, one demo event

2.3 API Project Structure

```
apps/api/
  Program.cs              # Minimal hosting, service registration
  appsettings.json        # DB, JWT, SendGrid, Stripe (empty Phase
1)
  Data/
    ApplicationDbContext.cs
    Migrations/
  Features/               # Vertical slice per feature area
    Auth/
    Events/
    Teams/
    Players/
    Scores/
    Sponsors/
    Emails/
    QR/
  Common/
    Middleware/            # Exception handler, JWT validation
Extensions/            # IServiceCollection, IApplicationBuilder
```

3. Organization & Event Setup

3.1 Organization Management

- POST /api/v1/auth/register — creates user + organization in one call. Returns JWT pair.
- GET/PATCH /api/v1/orgs/{id} — org name, slug, logo upload, mission statement, is_501c3, theme tokens
- Logo upload: POST /api/v1/orgs/{id}/logo — multipart/form-data. API resizes to 200px height, stores on CDN, returns URL.
- Theme save: validateContrast(primary, surface) from packages/theme runs server-side before saving. Returns 422 if contrast < 4.5:1.

3.2 Event Creation

- POST /api/v1/events — creates event under org. Returns event_code (8-char random alphanumeric).
- Format: scramble | stroke | stableford | best_ball
- Start type: shotgun (all teams start simultaneously on different holes) | tee_times (sequential)
- Holes: 9 or 18. All downstream calculations adapt.
- Shotgun setup: POST /api/v1/events/{id}/shotgun-assignments — auto-assigns teams to holes, validates no duplicates.
- Tee time setup: POST /api/v1/events/{id}/tee-times — start time + interval in minutes. Returns full schedule.

- Course setup: POST /api/v1/events/{id}/course — attach course record, set par and handicap_index per hole.

3.3 Hole Challenge Configuration

- POST /api/v1/events/{id}/challenges — create challenge: type, hole_number (null = all-day challenge), description, prize, sponsor_id
- PATCH /api/v1/challenges/{id}/result — admin enters result manually (distance in yards, player name). Phase 6 upgrades to GPS measurement.

4. Three-Mode Registration System

All three registration modes are built and enabled in Phase 1. Each mode can be independently toggled per event via events.config.

4.1 Mode 1 — Full Team Registration

- POST /api/v1/events/{id}/register/team — captain provides team name, all player details, and entry fee method
- Invite links (optional): POST /api/v1/teams/{id}/invite — generates a signed invite_token (HMAC, 48hr expiry). Sends email with registration link.
- Teammate self-registers via /e/{orgSlug}/{eventCode}/join?token={invite_token}
- Slot tracking: teams.max_players (default 4). When all slots filled, team.is_complete flips to true automatically.

4.2 Mode 2 — Join an Existing Team

- GET /api/v1/events/{id}/teams?q={search} — fuzzy name search returns open teams with available slots
- POST /api/v1/events/{id}/register/join — { team_id, player details }. Validates slot availability with optimistic lock.
- Captain receives email notification when a new player joins their team.

4.3 Mode 3 — Free Agent Pool

- POST /api/v1/events/{id}/register/free-agent — player details + preferences: skill_level (beginner|intermediate|advanced|competitive), age_group (under30|30to50|over50), gender, pairing_note
- GET /api/v1/events/{id}/free-agents — admin view: all unassigned players with preference cards
- POST /api/v1/events/{id}/free-agents/assign — { player_id, team_id }. Admin drag-and-drop. Triggers team assignment email.
- POST /api/v1/events/{id}/free-agents/auto-pair — handicap snake draft algorithm. Returns proposed pairings for admin review before committing.

- Auto-pair algorithm: sort free agents by handicap, snake-draft into groups of max_players. Soft preference matching (same age group) applied as tiebreaker.

5. Admin Dashboard (Expo Router Web)

The admin dashboard is a web app built with Expo Router. It shares ScoreCard, TeamCard, and other components from packages/ui with the Phase 2 mobile app. Organizers access it from any browser on a tablet, laptop, or phone.

5.1 Dashboard Screens — Phase 1

- **Event Hub (/admin/events/{id})** — Central screen: quick stats (teams checked in, scores complete, $ raised), nav to all panels below
- **Registration Panel** — Tabs: Complete Teams | Partial Teams | Free Agents | Pending Invites. Each team card shows players, payment status, and check-in status.
- **Free Agent Board** — Kanban-style: Unassigned pool on left, team cards on right. Drag-and-drop assignment. Auto-pair button with preview modal.
- **Check-In Screen** — Full-screen QR scanner (device camera) or name search. Individual and team check-in. Live counter widget.
- **Score Entry** — Select team → enter hole-by-hole gross scores. Running to-par displayed. Save auto-triggers leaderboard recalculation.
- **Leaderboard (Admin)** — Sortable standings table: rank, team, to-par, gross, thru. Not publicly accessible in Phase 1.
- **Hole Challenges** — Manual result entry per challenge. Closest-to-pin: enter distance. Longest drive: enter distance. Puts result on challenge leaderboard.
- **Sponsors Panel** — Upload logos, set tier and placements. Preview how sponsor appears on landing page and (in Phase 2) leaderboard.
- **Fundraising Panel** — Entry fees collected vs. expected. Donation log. Grand total counter.
- **Print Kit** — Button: Generate QR Kit PDF. Button: Generate Paper Scorecards PDF (one per team, pre-filled with hole, par, handicap_index).

6. QR Check-In

- Every registered player receives a unique QR code in their confirmation email. The QR encodes a signed token: { type: player_checkin, player_id, event_id, signature }
- Admin check-in screen: camera streams to QR decoder. On successful scan, PATCH /api/v1/players/{id}/checkin called. Player card flips to checked-in state with timestamp.
- Team check-in shortcut: if all players in a team are checked in, team.check_in_status flips to complete automatically.
- Walk-up registration: from check-in screen, admin taps 'Add Walk-Up'. Abbreviated form (name, email, team assignment). Generates and displays QR on screen for player to screenshot.
- Live counter component subscribes to check-in events (polling in Phase 1; SignalR in Phase 3). Shows 'X of N teams checked in'.

7. Automated Emails

SendGrid is configured in Phase 1. All email templates live in the email_templates table, are org-branded, and are sent by the Email feature in apps/api.

Trigger	Recipient	Content
Registration confirmation	All players on team	Event details, team name, unique QR check-in code, entry fee receipt, event date/location
Invite link	Invited teammate	Captain's name, team name, registration link with token, deadline
Team assignment	Free agent	Team name, captain name, starting hole or tee time, teammate roster
Event reminder	All checked-in players	24 hours before: tee time, starting hole, course address link, weather check suggestion
Thank-you	All players	Final score, leaderboard position, total funds raised, sponsor acknowledgments, donation CTA
Donation receipt	Donor	Amount, event name, org name, 501c3 note if is_501c3=true, tax year

Email Timing: Confirmation sends immediately on registration. Reminder sends via a scheduled job at event.start_at - 24 hours. Thank-you sends when admin marks event status = completed. Use Hangfire (background jobs library for .NET) for scheduled sends.

8. Public Landing Page (Next.js)

- Route: /e/{orgSlug}/{eventCode} — SSR rendered. Available in Phase 1 as a registration page. Leaderboard link appears in Phase 2.
- Above fold: org logo, event name, date, course name. Org theme colors applied via CSS custom properties.
- Registration CTAs: three buttons — Register a Team, Join a Team, I Need a Team. Each links to the relevant registration form.
- Donation widget: DonationForm component with amount presets and custom amount. POST /api/v1/events/{id}/donations. Thermometer bar shows progress toward goal.
- Sponsor showcase: logos displayed in tier order (Title → Gold → Hole → Silver → Bronze).
- Remaining slots counter: 'X team spots remaining' updated server-side on each page load (ISR, 60-second revalidate).
- JSON-LD Event structured data in <head> for Google Search rich results.
- Open Graph meta tags: event name, date, org logo — for social media sharing.

- 'Leaderboard Coming Soon' placeholder shown in Phase 1. Replaced with live leaderboard link in Phase 2.

9. Phase 1 API Endpoints

Method	Endpoint	Description
POST	/auth/register	Create organizer account + organization. Returns JWT pair.
POST	/auth/login	Validate credentials. Returns JWT pair.
POST	/auth/refresh	Exchange refresh token. Returns new JWT pair.
GET	/orgs/{id}	Get org details including theme and branding
PATCH	/orgs/{id}	Update org name, mission, theme, 501c3 flag
POST	/orgs/{id}/logo	Upload org logo (multipart). Returns CDN URL.
POST	/events	Create event under org. Returns event with generated event_code.
GET	/events/{id}	Full event config including course, holes, challenges
PATCH	/events/{id}	Update event settings, format, status
POST	/events/{id}/shotgun-assignments	Auto-assign teams to holes for shotgun start
POST	/events/{id}/tee-times	Generate tee time schedule from start time + interval
POST	/events/{id}/course	Attach course, set par and handicap_index per hole
POST	/events/{id}/register/team	Mode 1: full team registration
POST	/events/{id}/register/join	Mode 2: individual joins existing team
POST	/events/{id}/register/free-agent	Mode 3: solo player enters free agent pool

GET	/events/{id}/free-agents	Admin: list unassigned free agents with preferences
POST	/events/{id}/free-agents/assign	Admin: assign free agent to team. Triggers email.
POST	/events/{id}/free-agents/auto-pair	Admin: run handicap snake draft. Returns proposed pairings.
POST	/teams/{id}/invite	Generate signed invite token. Sends invite email.
PATCH	/players/{id}/checkin	Mark player as checked in. Updates team status.
PATCH	/scores/{teamId}/{holeNumber}	Admin score entry. Triggers leaderboard recalculation.
GET	/events/{id}/leaderboard	Admin-only standings (auth required in Phase 1)
CRUD	/events/{id}/sponsors	Sponsor management: logo, tier, placements
CRUD	/events/{id}/challenges	Hole challenge setup
PATCH	/challenges/{id}/result	Admin enters challenge result manually
GET	/events/{id}/fundraising	Totals: fees collected, donations, grand total
POST	/events/{id}/donations	Record a donation (manual or form submit)
GET	/events/{id}/qr/kit	Generate QR print kit (all event QR codes as ZIP)
GET	/events/{id}/scorecards/pdf	Paper backup scorecards PDF for all teams
GET	/pub/events/{eventCode}	Public: landing page data (no auth required)

10. What Phase 1 Deliberately Excludes

Not a gap: These are intentional deferrals. The database schema, API structure, and component library are designed for all of them. They're enabled in later phases, not retrofitted.

Excluded Feature	Added In	Why Deferred

Mobile scoring app (iOS/Android)	**Phase 2**	Admin scores manually. No mobile app yet.
Public leaderboard	**Phase 2**	Admin-only leaderboard ships Phase 1. Public launches with mobile app.
Offline scoring (SQLite)	**Phase 2**	No mobile app means no offline device.
QR scorecard transfer	**Phase 2**	Requires the mobile app.
Live SignalR updates	**Phase 3**	Admin dashboard uses polling in Phase 1. <2s push waits for Phase 3.
Email ad builder	**Phase 3**	Transactional emails ship Phase 1. Ad builder is a Phase 3 feature.
Stripe payments	**Phase 4**	Manual payment tracking in Phase 1. Online payment processing in Phase 4.
Auction (all formats)	**Phase 4**	Requires Stripe card-on-file.
League / handicap engine	**Phase 5**	Season tables exist in DB, populated Phase 5.
GPS / Mapbox	**Phase 6**	cup_location column exists, is null until Phase 6.